

2.4.1 Splitting Methods

Let $H(\mathbf{q}, \mathbf{p}) = H_1(\mathbf{q}, \mathbf{p}) + H_2(\mathbf{q}, \mathbf{p})$ have flow map $\mathcal{F}_h$, and indicate by $\mathcal{F}_h^1, \mathcal{F}_h^2$ the flow maps for the systems with Hamiltonians H_1, H_2 , respectively. The proposal is that the map

$$\mathcal{G}_h = \mathcal{F}_h^1 \circ \mathcal{F}_h^2$$

is an approximation of $\mathcal{F}_h$. For this to be a first order method, we need at least

$$\|\mathcal{G}_h(\mathbf{u}) - \mathcal{F}_h(\mathbf{u})\| \leq C(\mathbf{u})h^2.$$

Expand $\mathcal{F}_h(\mathbf{u})$ in a Taylor series:

$$\mathcal{F}_h(\mathbf{u}) = \mathbf{u} + h\mathbf{J}\nabla H + \mathcal{O}(h^2) = \mathbf{u} + h(\mathbf{J}\nabla H_1 + \mathbf{J}\nabla H_2) + \mathcal{O}(h^2). \quad (2.24)$$

On the other hand,

$$\mathcal{F}_h^1(\mathbf{u}) = \mathbf{u} + h\mathbf{J}\nabla H_1(\mathbf{u}) + \mathcal{O}(h^2), \quad \mathcal{F}_h^2(\mathbf{u}) = \mathbf{u} + h\mathbf{J}\nabla H_2(\mathbf{u}) + \mathcal{O}(h^2).$$

Composing the maps we have

$$\mathcal{F}_h^1 \circ \mathcal{F}_h^2(\mathbf{u}) = \mathbf{u} + h\mathbf{J}\nabla H_2(\mathbf{u}) + h\mathbf{J}\nabla H_1(\mathbf{u} + h\mathbf{J}\nabla H_2(\mathbf{u})) + \mathcal{O}(h^2).$$

Assuming H_1 is C^2 , we have

$$\mathcal{F}_h^1 \circ \mathcal{F}_h^2(\mathbf{u}) = \mathbf{u} + h(\mathbf{J}\nabla H_2(\mathbf{u}) + \mathbf{J}\nabla H_1(\mathbf{u})) + \mathcal{O}(h^2),$$

which agrees with the expansion (2.24) of the flow map through the terms of first order. Thus the splitting method does indeed provide a second order local approximation of the flow map.

Example 2.4 Let

$$H_1(\mathbf{q}, \mathbf{p}) = \mathbf{p}^T \mathbf{M}^{-1} \mathbf{p} / 2, \quad H_2(\mathbf{q}, \mathbf{p}) = U(\mathbf{q}),$$

then a splitting method for $H = H_1 + H_2$ may be obtained by determining the flow maps for each of the two parts. For H_1 we have the differential equations

$$\dot{\mathbf{q}} = \mathbf{M}^{-1} \mathbf{p}, \quad \dot{\mathbf{p}} = \mathbf{0}.$$

The second equation tells us that $\mathbf{p}$ is constant, so it is fixed at its initial value, whereas the first equation says that $\mathbf{q}$ evolves on a linear path, hence its flow map is

$$\begin{aligned} \mathbf{Q} &= \mathbf{q} + h\mathbf{M}^{-1} \mathbf{p}, \\ \mathbf{P} &= \mathbf{p}. \end{aligned}$$

Similarly, the flow map of the system with Hamiltonian $H_2 = U$ is

$$\begin{aligned}\mathbf{Q} &= \mathbf{q}, \\ \mathbf{P} &= \mathbf{p} - h\nabla U(\mathbf{q}).\end{aligned}$$

The composition of these maps simplifies to

$$\begin{aligned}\mathbf{Q} &= \mathbf{q} + h\mathbf{M}^{-1}\mathbf{P}, \\ \mathbf{P} &= \mathbf{p} - h\nabla U(\mathbf{q}),\end{aligned}$$

which is precisely the Symplectic Euler method. Similarly, composing the same two maps in the opposite order gives the adjoint Symplectic Euler method.

Example 2.5 (Verlet is Symplectic) For the Symplectic Euler method $\mathcal{G}_h$ and its adjoint method $\mathcal{G}_h^\dagger$, consider the composition

$$\mathcal{K}_h := \mathcal{G}_{h/2}^\dagger \circ \mathcal{G}_{h/2}.$$

Let

$$\begin{bmatrix} \bar{\mathbf{q}} \\ \bar{\mathbf{p}} \end{bmatrix} = \mathcal{G}_{h/2} \begin{bmatrix} \mathbf{q} \\ \mathbf{p} \end{bmatrix},$$

i.e.,

$$\bar{\mathbf{q}} = \mathbf{q} + (h/2)\mathbf{M}^{-1}\bar{\mathbf{p}}, \quad \bar{\mathbf{p}} = \mathbf{p} - (h/2)\nabla U(\mathbf{q}).$$

and set

$$\begin{bmatrix} \mathbf{Q} \\ \mathbf{P} \end{bmatrix} = \mathcal{G}_{h/2}^\dagger \begin{bmatrix} \bar{\mathbf{q}} \\ \bar{\mathbf{p}} \end{bmatrix},$$

i.e.,

$$\mathbf{Q} = \bar{\mathbf{q}} + (h/2)\mathbf{M}^{-1}\bar{\mathbf{p}}, \quad \mathbf{P} = \bar{\mathbf{p}} - (h/2)\nabla U(\mathbf{Q}).$$

The composition simplifies to

$$\begin{aligned}\bar{\mathbf{p}} &= \mathbf{p} - \frac{h}{2}\nabla U(\mathbf{q}), \\ \mathbf{Q} &= \mathbf{q} + h\mathbf{M}^{-1}\bar{\mathbf{p}}, \\ \mathbf{P} &= \bar{\mathbf{p}} - \frac{h}{2}\nabla U(\mathbf{Q}).\end{aligned}$$